

Part I

SUMMARY of Rules of Differentiation

Summary

February 21, 2026

Summary

- **The Power Rule:** For any real number n , $\frac{d}{dx}x^n = nx^{n-1}$.
- **The Natural Exponential Function:** $\frac{d}{dx}e^x = e^x$.
- **The Sine Function:** $\frac{d}{dx}\sin(x) = \cos(x)$.
- **The Sum Rule:** If $f(x)$ and $g(x)$ are differentiable functions and c is a constant, then:

$$- \frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x),$$

$$- \frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x),$$

$$- \frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x).$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 For each of the following functions, use the "short cut derivative rules" to compute their derivative.

(a) $f(x) = \sqrt{x}$

Explanation.

$$f(x) = \sqrt{x} = x^{\frac{1}{2}}.$$

$$\begin{aligned} f'(x) &= \frac{1}{2}x^{\frac{1}{2}-1} \\ &= \frac{1}{2}x^{-\frac{1}{2}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

(b) $s(u) = \frac{5}{u^2}$

Explanation.

$$s(u) = \frac{5}{u^2} = 5u^{-2}.$$

$$\begin{aligned} s'(u) &= 5(-2)u^{-2-1} \\ &= -10u^{-3} \\ &= \frac{-10}{u^3} \end{aligned}$$

(c) $p(t) = t^5 + 4t^3 + \pi$

Explanation.

$$\begin{aligned} p'(t) &= 5t^{5-1} + 4(3)t^{3-1} + 0 \\ &= 5t^4 + 12t^2 \end{aligned}$$

Note that $\frac{d}{dx}(\pi) = 0$ because π is a constant.

Problem 2

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Problem 2 Given the polynomial function q defined by $q(v) = 2v^3 - 5v^2 + 7v - 9$ find:

- (a) The slope of the tangent line to the graph of q at the point where $v = 3$ using the limit definition of a derivative.

Explanation. To compute $q'(3)$ from the limit definition we use either

$$\lim_{v \rightarrow 3} \frac{q(v) - q(3)}{v - 3} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{q(3 + h) - q(3)}{h}.$$

Using the second formula:

$$\begin{aligned} q'(3) &= \lim_{h \rightarrow 0} \frac{q(3 + h) - q(3)}{h} \quad \text{which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{(2(3 + h)^3 - 5(3 + h)^2 + 7(3 + h) - 9) - (54 - 45 + 21 - 9)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2(27 + 27h + 9h^2 + h^3) - 5(9 + 6h + h^2) + 21 + 7h - 9) - 21}{h} \\ &= \lim_{h \rightarrow 0} \frac{54 + 54h + 18h^2 + 2h^3 - 45 - 30h - 5h^2 + 21 + 7h - 9 - 21}{h} \\ &= \lim_{h \rightarrow 0} \frac{31h + 13h^2 + 2h^3}{h} \\ &= \lim_{h \rightarrow 0} (31 + 13h + 2h^2) \\ &= 31. \end{aligned}$$

- (b) The slope of the tangent line to the graph of q at the point where $v = 3$ using the “short-cut derivative rules” to find a formula for q' and evaluating $q'(3)$.

Explanation.

$$\begin{aligned} q'(v) &= (2v^3 - 5v^2 + 7v - 9)' \\ &= (2v^3)' + (-5v^2)' + (7v)' + (-9)' \\ &= 3 \cdot 2 \cdot v^{3-1} + -5 \cdot (2) \cdot v^{2-1} + 7 \cdot 1 \cdot v^{1-1} + 0 \\ &= 6v^2 - 10v + 7 \\ &\implies q'(3) = 6 \cdot 3^2 - 10 \cdot 3 + 7 = 31. \end{aligned}$$

- (c) The equation of the tangent line to the graph of q at $v = 3$.

Explanation. Slope of tangent line to the graph of q at the point where $v = 3$: $q'(3) = 6 \cdot 3^2 - 10 \cdot 3 + 7 = 31$.

Point on tangent line at the point where $v = 3$: $(3, q(3)) = (3, 21)$.

Problem 2

Equation of tangent line to the graph of q at the point where $v = 3$:

$$\begin{aligned}y - 21 &= 31(x - 3) \\ \implies y &= 31x - 72.\end{aligned}$$

Problem 3

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Problem 3

Problem 3 Find s' of the function s defined by $s(t) = 3t^2 + 5e^t - \frac{1}{t}$.

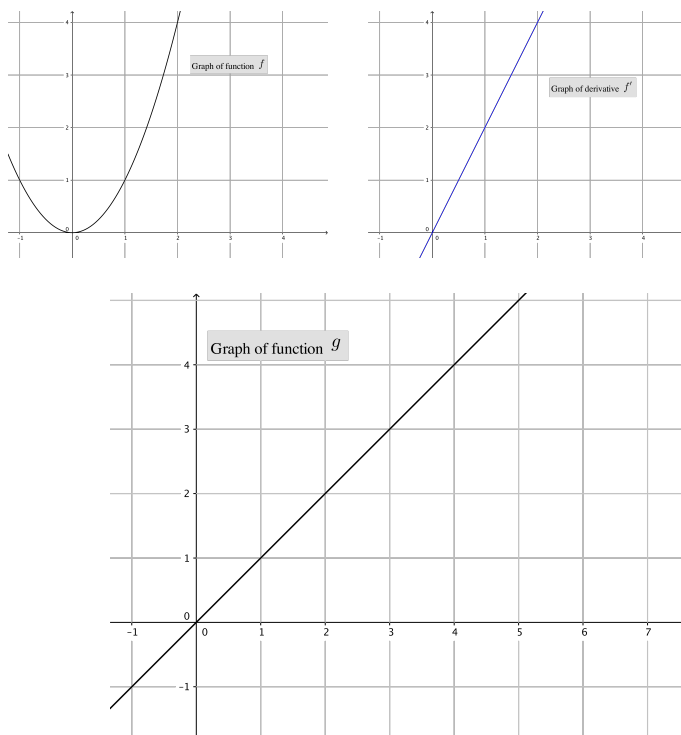
Explanation. $s'(t) = 6t + 5e^t + \frac{1}{t^2}$.

Problem 4

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Problem 4

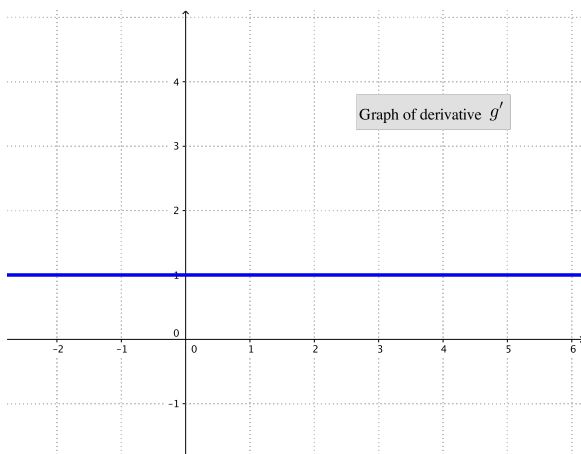
Problem 4 Use the given graphs of f and g and their accompanying derivatives to answer the following questions.



- (a) Write an equation for the tangent line to f at $x = 2$.

Explanation. $f(2) = 4$, $f'(2) = 4$, $y - 4 = 4(x - 2)$, $y = 4x - 4$

- (b) Draw the graph of g'



Explanation.

- (c) Compute the value of the $(5f + 3g)'(2)$.

Explanation.

$$\begin{aligned}(5f + 3g)'(2) &= 5 \cdot f'(2) + 3 \cdot g'(2) \\ &= 5 \cdot 4 + 3 \cdot 1 \\ &= 20 + 3 = 23.\end{aligned}$$

- (d) Find the equation of the tangent line to the graph of $(5f + 3g)$ at the point where $x = 2$.

Explanation.

$$\begin{aligned}(5f + 3g)(2) &= 5 \cdot f(2) + 3 \cdot g(2) \\ &= 5 \cdot 4 + 3 \cdot 2 \\ &= 20 + 6 = 26\end{aligned}$$

$$\begin{aligned}y - 26 &= 23(x - 2) \\ y &= 23x - 20\end{aligned}$$

- (e) Use the given graph of f' and g' to find the following:

- (i) A formula for f'

Explanation. $f'(x) = 2x$. The graph shows a linear function through the point $(0, 0)$ and $(1, 2)$.

- (ii) A formula for g'

Explanation. $g'(x) = 1$. The graph shows a linear function with slope 0 and through the point $(0, 1)$.

- (f) Find the expression for $(5f + 3g)'(x)$.

Explanation.

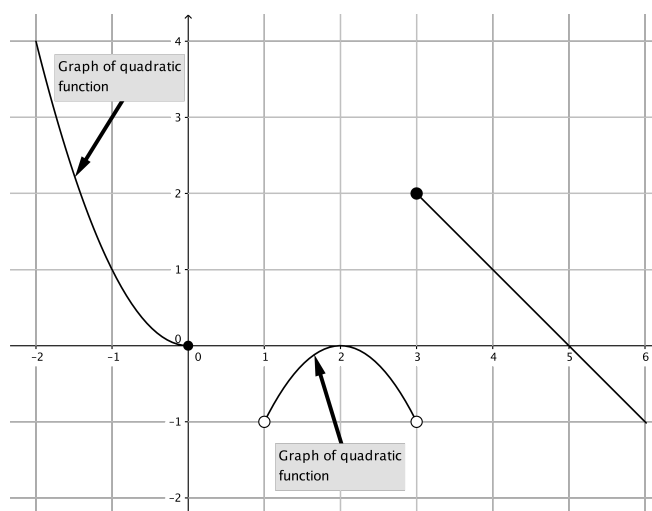
$$\begin{aligned}(5f + 3g)'(x) &= 5 \cdot f'(x) + 3 \cdot g'(x) \\ &= 5 \cdot 2x + 3 \cdot 1 \\ &= 10x + 3\end{aligned}$$

Problem 5

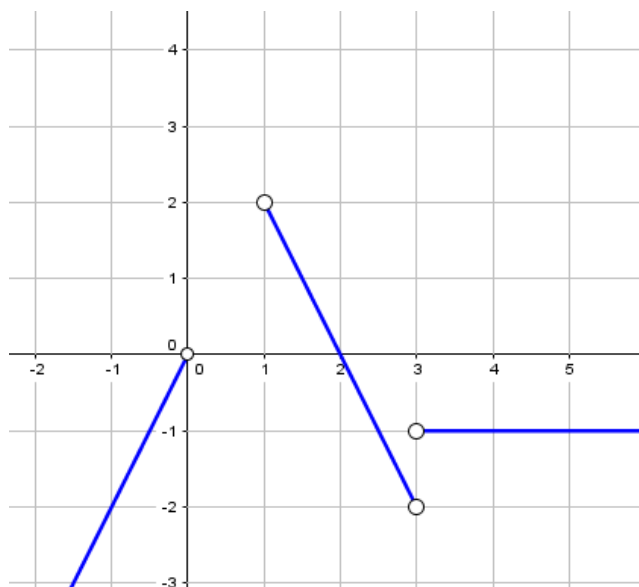
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Problem 5

Problem 5 Sketch the graph of the derivative of the given function:



Explanation.



Problem 6

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Problem 6 A company is producing cell phones. The cost of producing x cell phones is given by $C(x)$, defined by

$$C(x) = -0.01x^2 + 40x + 400, \text{ for } 0 \leq x < 1000.$$

AVERAGE cost of producing the first x cell phones is given by

$$C_{AVG} = \frac{C(x)}{x}$$

If the company has produced x cell phones, the cost of producing one more item is given by

$$\text{COST of producing one more item} = C(x+1) - C(x)$$

MARGINAL COST is approximation of the cost of producing one more cell phone

$$\text{MARGINAL COST} = C'(x)$$

- (a) Compute the average cost of the first 300 cellphones that the company produces.

Explanation. The first 300 cell phones costs, on average,

$$C_{AVG} = \frac{C(300)}{300} = 38.33$$

- (b) Compute the cost of producing one more cell phone, if the company has produced 300 cell phones.

Explanation.

$$C(301) - C(300) = -0.01(301)^2 + 40(301) + 400 - (-0.01(300)^2 + 40(300) + 400) = 33.99$$

- (c) Compute the marginal cost, if 300 cell phones have been produced.

Explanation.

$$\text{MARGINAL COST} = C'(300) = [0.02x + 40]_{x=300} = 34$$

- (d) Why is the marginal cost a good approximation of the cost of producing one more item? Explain!

Explanation. By the definition of the derivative we have that

$$C'(300) = \lim_{h \rightarrow 0} \frac{C(300+h) - C(300)}{h} \approx \frac{C(300+h) - C(300)}{h}$$

Problem 6

for any h near 0. In particular, for $h = 1$, we obtain

$$C'(300) \approx \frac{C(300+1) - C(300)}{1} = C(301) - C(300),$$

Therefore, marginal cost at $x = 300 \approx$ the cost of producing one more item!